YZM5110 — MULTIMODAL LEARNING

RESEARCH PROJECT REPORT

INVESTIGATING THE EFFECT OF PROMPT DEPTH ON PARAMETER EFFICIENCY AND UNSEEN-CLASS GENERALIZATION IN VISION-LANGUAGE MODELS

MaPLe · CLIP ViT-B/16 · EuroSAT RGB

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Public repository: https://github.com/siyahkut/yzm5110_research_1

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ABSTRACT

This study studies how coupled prompt depth and regularization of trainable deep text prompts affect few-shot base-to-novel generalization and parameter efficiency in Multi-modal Prompt Learning (MaPLe). A frozen CLIP ViT-B/16 is evaluated on EuroSAT RGB with 16 labeled training examples per base class, five epochs and three seeds. First, a five-depth MaPLe ablation (K=1,3,5,7,9) finds the strongest observed baseline at K=7 (mean seed-wise harmonic mean [HM] 81.365%) rather than K=9 (76.086%). We then implement an additional loss term on compound text prompts: DAPR (Deep-prompt regularization with depth-dependent weights). At K=9, uniform weighting (gamma=0, lambda=0.1) achieves 79.828% HM, compared with 77.703% and 77.909% for depth-weighted gamma=1 and gamma=2. Lambda ablation at K=9 yields 73.623% (0.01), 79.828% (0.1), and 76.042% (0.5). At K=7, uniform regularization (gamma=0, lambda=0.1) obtains 82.134% HM, 0.769 percentage points above the local K=7 baseline. These findings suggest that prompt regularization can improve the observed base-to-novel balance under selected settings; they do not show a general advantage of depth-aware weighting over uniform weighting. Results are exploratory, since choices were informed by test-set evaluations on a single dataset with three seeds. Full cross-dataset confirmation, paired significance tests, and standardised resource measurements remain out of scope.
Keywords: MaPLe; DAPR; deep-prompt regularization; prompt depth; base-to-novel transfer; EuroSAT; parameter efficiency.

Keywords: MaPLe; CLIP; prompt depth; vision–language models; base-to-novel generalization; parameter efficiency.

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LIST OF ABBREVIATIONS AND SYMBOLS

CLIP Contrastive Language–Image Pre-training

MaPLe Multi-modal Prompt Learning

CoOp Context Optimization

CoCoOp Conditional Context Optimization

VPT Visual Prompt Tuning

HM Harmonic Mean

K Prompt depth (number of prompted transformer layers)

B Base-class accuracy

N Novel-class accuracy

SD Sample standard deviation

SGD Stochastic Gradient Descent

1. Research question, scope, motivation, and contributions

Research questions: (1) Under a fixed 16-shot budget, how does coupled prompt depth affect base accuracy, novel-class generalization, harmonic mean, and trainable parameter count? (2) Does adding L2 regularization to MaPLe deep text prompts improve base-to-novel performance at fixed depth? (3) Do depth-dependent penalties outperform uniform penalties, and how sensitive is performance to lambda?

We expected that deeper prompts would give the model more trainable parameters. This may improve performance on base classes, but it may not help with novel classes. We also tested whether a penalty on deep prompts could improve this balance. The experiment compares several depth settings while keeping the training setup the same.

The project has five main contributions. First, we tested five prompt depths with three seeds each. Second, we added regularization to deep text prompts in trainers/maple.py without changing the frozen CLIP encoders. Third, we tested gamma values 0, 1 and 2 and lambda values 0.01, 0.1 and 0.5. Fourth, we compared the method with MaPLe at depths 7 and 9. Finally, we calculated the results from correct prediction counts and reported uncertainty and limitations.

2. Literature search approach and critical synthesis

We selected five related methods to understand how prompt learning developed. CLIP provides image and text representations. CoOp learns text prompts, CoCoOp creates image-dependent text prompts, and VPT learns visual prompts. MaPLe combines prompts in both branches. We used official publications to compare their goals, training methods and limitations. This is a focused literature review, not a systematic review.

Table 1 compares five key papers in one place. The published results from MaPLe are discussed in Section 4. They are not our own test results. We studied CLIP, CoOp, CoCoOp and VPT in the literature but did not run their experiments.

Paper	Main idea	Input / assumptions	How it works	Data / tests	Comparison in this study	Code / resources	Main limit
CLIP [1]	Learn general image–text links	Images and text; class names at test time	Two encoders trained to match images with text	Large image–text pretraining	Used as the frozen starting model	Public code and models; local weight hash not recorded	Can be sensitive to wording and new domains
CoOp [2]	Learn text prompts	Labeled images and class names; frozen CLIP	Updates learned text context using classification loss	Few-shot base/novel tests in the paper	Related text-only approach; not rerun	Public code; exact local version not checked	May fit seen classes too closely
CoCoOp [3]	Create prompts based on each image	Input image; frozen CLIP	Uses an image to build a text prompt	Tests transfer to new classes	Alternative method; not rerun	Needs extra model parts and compute	Adds model complexity and runtime cost
VPT [4]	Learn prompts for the image encoder	Images; frozen vision transformer	Adds learnable visual tokens	Image recognition and transfer benchmarks	Visual-only comparison; not rerun	Code available; new setup needed	Does not directly connect text and image prompts
MaPLe [5]	Learn linked text and visual prompts	Images, class names, frozen CLIP	Learned text prompts and text-to-vision projections	Our EuroSAT 16-shot base/novel experiments	Main baseline; five depths, three seeds	Our logs, configs and checkpoints; commit needs checking	Higher depth may not help novel classes

Together, these papers show how prompts can adapt a frozen model. CoOp changes text prompts, VPT changes visual inputs, CoCoOp makes text prompts depend on the image, and MaPLe learns linked text and image prompts. Our question is whether deeper prompts are worth their extra trainable parameters and whether regularization helps novel-class accuracy.

3. Technical explanation of the selected method

CLIP maps an image x to a visual representation $v=f_v(x)$ and each class prompt t_c to a text representation $u_c=f_t(t_c)$. The class logit is proportional to cosine similarity: $z_c = \exp(\text{logit_scale}) \times (v \cdot u_c)/(\|v\| \|u_c\|)$. Supervised cross-entropy on the labeled base examples updates prompt-related parameters; the pretrained visual and textual encoder weights remain frozen.

MaPLe learns shallow text context tokens and a projection that couples these to visual prompts. Deeper levels introduce compound text prompts with associated visual projections. With context length $N_{CTX}=2$, text width 512 and visual width 768, one prompt level contributes $2\times512 + (512\times768 + 768) = 395{,}008$ trainable scalar parameters. Under this implementation, D levels yield $P_{train}(D) = 395{,}008 \times D$.

For D=7: shallow/compound text context across seven levels contributes $7\times1{,}024 = 7{,}168$ parameters, while seven projection layers contribute $7\times393{,}984 = 2{,}757{,}888$; sum = 2,765,056. Frozen parameters remain 124,323,841, hence $P_{total}(7)=127{,}088{,}897$.

4. Reproduction target, implementation, and deviations

Targeted testable claim: increasing MaPLe prompt depth does not necessarily yield monotonically better novel-class transfer or a better base/novel balance. A strictly increasing novel accuracy and harmonic mean across depths would contradict this specific hypothesis. The experiment adapts the publicly described MaPLe method using the local multimodal-prompt-learning implementation, with the same backbone and fixed training configuration across depth conditions.

This is a reduced-scope reproduction of the published EuroSAT depth-9 setting plus a controlled depth ablation, NOT a full replication of the 11-dataset benchmark. The paper states ViT-B/16 CLIP, 16 shots per base class, prompt depth 9, prompt lengths of 2, 5 epochs, batch size 4, SGD with learning rate 0.0035, and results averaged across three runs. These overlap with the local configuration. Key scope and provenance differences are single-dataset evaluation instead of 11, a laptop RTX 5070 Ti GPU rather than an A100, depth changes, local dependency adjustments, and missing verified source commit / dataset checksum / checkpoint hash. Do not attribute score discrepancies to prompt depth without eliminating possible split/version/implementation differences.

4.1 Original published EuroSAT scores versus our measured reference

Published scores below are copied from Khattak et al. (CVPR 2023), Table 3, EuroSAT panel (k), page 19119. They are the authors' published results, not outputs of our local runs. The reference configuration in our study is depth 9, chosen because the paper lists MaPLe prompt depth 9 as its main setting.

Published method (Table 3)	Base Acc. %	Novel Acc. %	HM %
CLIP (zero-shot)	56.48	64.05	60.03
CoOp	92.19	54.74	68.69
Co-CoOp	87.49	60.04	71.21
MaPLe (published)	94.07	73.23	82.35

Our depth-9 MaPLe reference measured across seeds 1, 2 and 3: Base = 93.579 ± 0.805%; Novel = 64.179 ± 4.216%; mean of per-seed HM = 76.086 ± 2.957 percentage points (mean ± sample SD). The descriptive differences, local minus published MaPLe, are Base = 93.579−94.070 = −0.491 pp; Novel = 64.179−73.230 = −9.051 pp; HM = 76.086−82.350 = −6.264 pp. These are *cross-protocol descriptive gaps*, NOT a fair controlled hypothesis test; paper table aggregates and the mean of per-seed harmonic means may also differ in convention. The large novel gap remains an important reproduction discrepancy requiring investigation of exact split membership, setup revision, and checkpoint provenance.

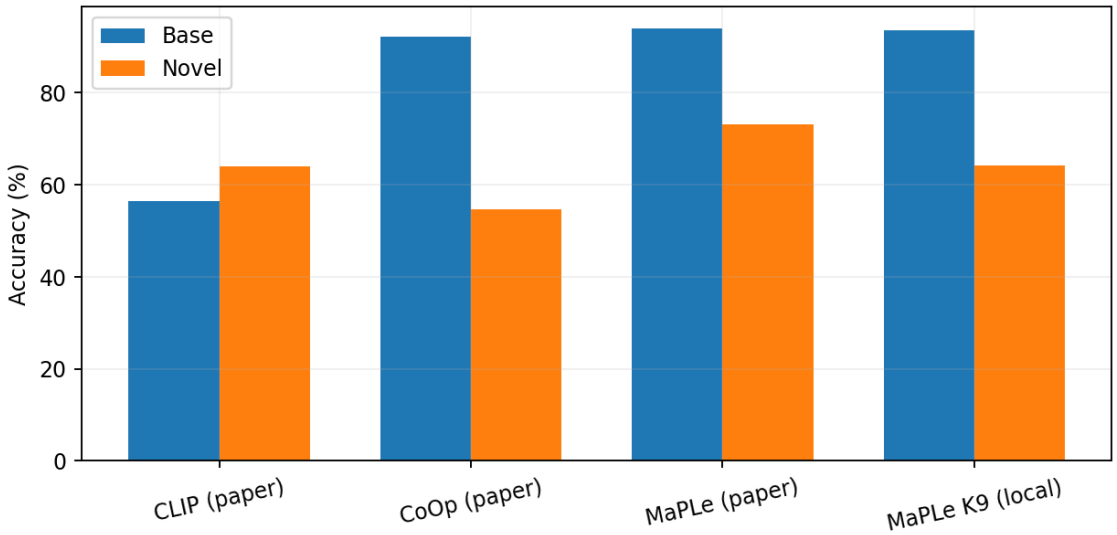


Figure 1. Published MaPLe EuroSAT results (CVPR 2023, Table 3) versus local depth-9 three-seed means. Cross-protocol descriptive comparison, not a controlled benchmark replication.

Numerical audit: 64.1795 − 73.2300 = −9.0505 percentage points in novel accuracy; 93.5794 − 94.0700 = −0.4906 pp in base accuracy; 76.0859 − 82.3500 = −6.2641 pp in HM. Importantly, the local HM shown here is an arithmetic mean of three individually calculated seed-level harmonic means. The original paper may aggregate over repetitions differently; a comparison of displayed HM values is so descriptive, not a significance test.

The original paper's global, 11-dataset average is 82.28% Base / 75.14% Novel / 78.55% HM for MaPLe (Table 3, average panel), which is a different aggregation target and must NOT be compared as a EuroSAT-specific measurement.

Local compatibility adaptations included removing the obsolete verbose argument from the Dassl learning-rate scheduler constructor and using torch.load(..., weights_only=False) for locally trusted checkpoints. The latter is unsafe for untrusted checkpoints and should not be applied to arbitrary

downloaded files. Trainable-parameter instrumentation was added after freezing encoders and before loading the checkpoint.

4.2 Reproduction discrepancy: observations, hypotheses, and checks

Observed outcome (not a hypothesis): published EuroSAT novel-class accuracy is 73.23%, while the local depth-9 evaluation yields 68.718%, 60.385%, and 63.436% on seeds 1, 2, and 3. Their mean is 64.179%, their sample SD is 4.216 pp, and their max–min spread is 8.333 pp. Base accuracy remains close to the published number. The incomplete novel-class reproduction should not be concealed by the stronger performance of a separately tested depth-7 variant.

Potential explanations should be distinguished from established causes. Different few-shot sample selections and random initializations can affect prompt adaptation. Split membership, class-name mappings, image files, CLIP text template wording, and preprocessing must be checked at artifact level. Dependency changes, pretrained weight provenance, code revision, and numerical precision could matter, but no available evidence isolates any one as the source of the deficit. Hardware differences, particularly, are not by themselves proof of poorer generalization.

Factor	Evidence available	Unverified equivalence / diagnostic
Random seeds and few-shot samples	Three local seeds; novel scores vary by 8.333 pp	Published seed-specific selections and initialization are not available for a matched paired comparison.
Dataset / split	EuroSAT split_zhou_EuroSAT.json; 5+5 classes, 4,200/3,900 test images	Exact file hashes, membership, order, and class label/text mappings vs the source benchmark.
Preprocessing / prompts	Local RGB 224×224 transforms and MaPLe N_CTX=2 recorded	Exact transform implementations, interpolation, text prompt strings, and original preprocessing artifacts.
Source code / checkpoint	Local epoch-5 depth-9 checkpoints load and evaluate	Original repository commit, CLIP weights hash, local checkpoint hashes, official pretrained model evaluation.
Runtime / precision	Windows 11, PyTorch 2.11.0+cu128, RTX 5070 Ti Laptop GPU	Differences from the published environment and any measurable impact under controlled replay.
Metric aggregation	Local HM computed seed-wise and then averaged	Whether the published Table 3 reports HM(mean base, mean novel) or mean of per-seed HM.

4.3 Prioritized protocol verification

Priority 1 — If authors' released checkpoints are accessible, evaluate the original depth-9 EuroSAT checkpoint on the local dataset and pipeline. A score near 73.23% would point toward local training/reproduction differences; a similarly low score would prompt scrutiny of dataset, class mapping, and evaluation. This is a proposed test; it has not been performed.

Priority 2 — Hash and diff split_zhou_EuroSAT.json, dataset image lists, class names, all relevant YAML configs, CLIP weights, and the source Git commit. Capture exact training and evaluation commands and checkpoint fingerprints. These checks are not yet complete.

Priority 3 — Export per-class predictions and a novel-class confusion matrix, then compare seed-specific failure modes. Preserve all three runs and study changes without selecting or tuning on the test set.

Priority 4 — For a confirmatory depth-7-versus-depth-9 claim, select hyperparameters on a separate validation protocol, freeze that decision, and evaluate on independent test data or an additional dataset. The present ranking is exploratory because the existing test results informed depth selection.

5. Proposed improvement and rationale

We changed the MaPLe training loss by adding a penalty to the learnable deep text prompts (compound_prompts_text). We did not change the pretrained CLIP encoder weights. The new loss equals classification loss plus lambda times a weighted L2 penalty. Gamma controls how much weight each prompt depth receives. When gamma is zero, all deep prompts receive equal weight.

We compared the method with the original MaPLe at the same prompt depth (lambda = 0). At depth 9, we used gamma = 0, 1 and 2 while keeping lambda = 0.1. We also tested lambda = 0.01, 0.1 and 0.5 with gamma = 0. Finally, we compared regularized and original MaPLe at depth 7. These tests help separate the effect of regularization from the effect of depth-based weights.

6. Experimental design, data, splits, baselines, metrics, resources

Component	Local experimental setting
Dataset	EuroSAT RGB; split_zhou_EuroSAT.json
Base / novel classes	5 base classes / 5 novel classes
Base train	16 shots per class × 5 = 80 images; 20 batches/epoch at batch size 4
Test	4,200 base images; 3,900 novel images
Backbone / trainer	CLIP ViT-B/16; MaPLe; frozen CLIP encoders
Prompt depths	1, 3, 5, 7, 9
Random seeds	1, 2, 3 for every depth
Training	5 epochs; SGD; LR 0.0035; batch 4; N_CTX 2
Evaluation	Base and novel subsets; accuracy and macro-F1
System	Windows 11; Python 3.10.20; PyTorch 2.11.0+cu128; RTX 5070 Ti Laptop GPU
Measured resources	Trainable/total parameter counts; elapsed values from logs
Unavailable resource metric	Peak GPU memory: not recorded; no value imputed

The reported class split follows the supplied EuroSAT split file; the actual per-class membership should be included in the final repository. Different seeds correspond to different few-shot samples/checkpoints. Accuracy is computed from correct-prediction counts. Macro-F1 is taken from the evaluator logs rather than reconstructed from a confusion matrix, because full per-class prediction counts were not supplied. DAPR extensions re-use the same five base classes and five novel classes (4,200 and 3,900 test examples respectively) with seed-dependent 16-shot subsets. For DAPR experiments, K is fixed by TRAINER.MAPLE.PROMPT_DEPTH. The environment variables DAPR_LAMBDA and DAPR_GAMMA define the additional loss term. The evaluated checkpoint is epoch 5 for each run.

7. Metrics and complete calculation method

Accuracy (percentage) = 100 × C/N, where C is the number of correct predictions and N the test-set size. Example, depth 7 seed 1: base = 100×3,872/4,200 = 92.190476%; novel = 100×2,804/3,900 = 71.897436%. Use unrounded values for downstream computations.

Per-seed harmonic mean (HM) = 2BN/(B+N), where B and N are the base and novel accuracies expressed in the SAME unit (both percentages or both proportions). HM penalizes an imbalance between base and novel performance. Example, depth 7 seed 1: HM = 2×(3872/4200×100)×(2804/3900×100) / [(3872/4200×100)+(2804/3900×100)].

Mean for a depth: $\bar{x} = (x_1+x_2+x_3)/3$. Sample standard deviation: $s = \sqrt{[(x_1-\bar{x})^2+(x_2-\bar{x})^2+(x_3-\bar{x})^2] / (3-1)}$. Reported ± values are sample SD, not standard error or a confidence interval. Harmonic mean is calculated separately for each seed and then averaged; in general, HM(mean base, mean novel) differs from mean(HM per seed). In the DAPR analysis, we average seed-wise HM values and compute sample SD over three HM values; we do not compute HM from mean base and mean novel. Differences are in percentage points and are descriptive only.

Macro-F1 = $(1/K)\ \Sigma_c\ [2\times Precision_c\times Recall_c/(Precision_c+Recall_c)]$, with the usual zero-division convention defined by the evaluator. The macro-F1 percentages in the tables were logged by the evaluation code. Without per-class confusion counts, exact independent reconstruction is not possible.

Generalization gap = Base accuracy − Novel accuracy in percentage points (pp). Trainable ratio (%) = 100×P_train/P_total. Relative trainable-parameter reduction of depth 7 vs 9 = 100×(P_train(9)−P_train(7))/P_train(9). Difference in mean HM = mean HM(7)−mean HM(9), in pp.

8. Complete seed-level results and arithmetic audit

Depth	Seed	Base C/4200	Novel C/3900	Base %	Novel %	HM %	Base F1	Novel F1
1	1	3397	2721	80.88	69.77	74.92	80.8	68.1
1	2	3398	1737	80.90	44.54	57.45	80.9	32.2
1	3	3245	2420	77.26	62.05	68.83	77.1	60.0
3	1	3654	2769	87.00	71.00	78.19	86.7	68.2
3	2	3693	2420	87.93	62.05	72.76	88.1	57.4

Depth	Seed	Base C/4200	Novel C/3900	Base %	Novel %	HM %	Base F1	Novel F1
3	3	3502	1852	83.38	47.49	60.51	83.0	44.5
5	1	3259	1876	77.60	48.10	59.39	75.7	42.8
5	2	3621	2143	86.21	54.95	67.12	86.2	51.1
5	3	3808	2322	90.67	59.54	71.88	90.7	54.5
7	1	3872	2804	92.19	71.90	80.79	92.2	69.3
7	2	3575	3000	85.12	76.92	80.81	85.2	76.2
7	3	3945	2868	93.93	73.54	82.49	94.0	64.9
9	1	3943	2680	93.88	68.72	79.35	93.7	66.3
9	2	3956	2355	94.19	60.38	73.59	94.2	54.7
9	3	3892	2474	92.67	63.44	75.31	92.7	59.6

8.1 Depth 1: all three harmonic-mean calculations

Seed 1: B = 100×3,397/4,200 = 80.880952%; N = 100×2,721/3,900 = 69.769231%; HM = (2×80.880952×69.769231)/(80.880952+69.769231) = 74.915300%.

Seed 2: B = 100×3,398/4,200 = 80.904762%; N = 100×1,737/3,900 = 44.538462%; HM = (2×80.904762×44.538462)/(80.904762+44.538462) = 57.450272%.

Seed 3: B = 100×3,245/4,200 = 77.261905%; N = 100×2,420/3,900 = 62.051282%; HM = (2×77.261905×62.051282)/(77.261905+62.051282) = 68.826223%.

8.2 Depth 3: all three harmonic-mean calculations

Seed 1: B = 100×3,654/4,200 = 87.000000%; N = 100×2,769/3,900 = 71.000000%; HM = (2×87.000000×71.000000)/(87.000000+71.000000) = 78.189873%.

Seed 2: B = 100×3,693/4,200 = 87.928571%; N = 100×2,420/3,900 = 62.051282%; HM = (2×87.928571×62.051282)/(87.928571+62.051282) = 72.757513%.

Seed 3: B = 100×3,502/4,200 = 83.380952%; N = 100×1,852/3,900 = 47.487179%; HM = (2×83.380952×47.487179)/(83.380952+47.487179) = 60.511695%.

8.3 Depth 5: all three harmonic-mean calculations

Seed 1: B = 100×3,259/4,200 = 77.595238%; N = 100×1,876/3,900 = 48.102564%; HM = (2×77.595238×48.102564)/(77.595238+48.102564) = 59.388945%.

Seed 2: B = 100×3,621/4,200 = 86.214286%; N = 100×2,143/3,900 = 54.948718%; HM = (2×86.214286×54.948718)/(86.214286+54.948718) = 67.119066%.

Seed 3: B = 100×3,808/4,200 = 90.666667%; N = 100×2,322/3,900 = 59.538462%; HM = (2×90.666667×59.538462)/(90.666667+59.538462) = 71.877091%.

8.4 Depth 7: all three harmonic-mean calculations

Seed 1: B = 100×3,872/4,200 = 92.190476%; N = 100×2,804/3,900 = 71.897436%; HM = (2×92.190476×71.897436)/(92.190476+71.897436) = 80.789118%.

Seed 2: B = 100×3,575/4,200 = 85.119048%; N = 100×3,000/3,900 = 76.923077%; HM = (2×85.119048×76.923077)/(85.119048+76.923077) = 80.813789%.

Seed 3: B = 100×3,945/4,200 = 93.928571%; N = 100×2,868/3,900 = 73.538462%; HM = (2×93.928571×73.538462)/(93.928571+73.538462) = 82.492208%.

8.5 Depth 9: all three harmonic-mean calculations

Seed 1: B = 100×3,943/4,200 = 93.880952%; N = 100×2,680/3,900 = 68.717949%; HM = (2×93.880952×68.717949)/(93.880952+68.717949) = 79.352399%.

Seed 2: B = 100×3,956/4,200 = 94.190476%; N = 100×2,355/3,900 = 60.384615%; HM = (2×94.190476×60.384615)/(94.190476+60.384615) = 73.590843%.

Seed 3: B = 100×3,892/4,200 = 92.666667%; N = 100×2,474/3,900 = 63.435897%; HM = (2×92.666667×63.435897)/(92.666667+63.435897) = 75.314498%.

9. Mean, sample SD, and uncertainty across seeds

Depth	Base mean ± SD (%)	Novel mean ± SD (%)	HM mean ± SD (%)	Trainable parameters
1	79.68 ± 2.10	58.79 ± 12.93	67.06 ± 8.86	395,008
3	86.10 ± 2.40	60.18 ± 11.87	70.49 ± 9.06	1,185,024
5	84.83 ± 6.65	54.20 ± 5.75	66.13 ± 6.30	1,975,040
7	90.41 ± 4.67	74.12 ± 2.56	81.37 ± 0.98	2,765,056
9	93.58 ± 0.81	64.18 ± 4.22	76.09 ± 2.96	3,555,072

Depth 1, Base accuracy: mean = (80.880952 + 80.904762 + 77.261905)/3 = 79.682540; squared deviations = [1.436192, 1.493827, 5.859472]; s = sqrt((1.436192 + 1.493827 + 5.859472)/2) = 2.096365 pp.

Depth 1, Novel accuracy: mean = (69.769231 + 44.538462 + 62.051282)/3 = 58.786325; squared deviations = [120.624224, 203.001600, 10.659944]; s = sqrt((120.624224 + 203.001600 + 10.659944)/2) = 12.928375 pp.

Depth 1, Harmonic mean: mean = (74.915300 + 57.450272 + 68.826223)/3 = 67.063932; squared deviations = [61.643985, 92.422452, 3.105671]; s = sqrt((61.643985 + 92.422452 + 3.105671)/2) = 8.864878 pp.

Depth 1, Base macro-F1: mean = (80.800000 + 80.900000 + 77.100000)/3 = 79.600000; squared deviations = [1.440000, 1.690000, 6.250000]; s = sqrt((1.440000 + 1.690000 + 6.250000)/2) = 2.165641 pp.

Depth 1, Novel macro-F1: mean = (68.100000 + 32.200000 + 60.000000)/3 = 53.433333; squared deviations = [215.111111, 450.854444, 43.121111]; s = sqrt((215.111111 + 450.854444 + 43.121111)/2) = 18.829321 pp.

Depth 3, Base accuracy: mean = (87.000000 + 87.928571 + 83.380952)/3 = 86.103174; squared deviations = [0.804296, 3.332073, 7.410494]; s = sqrt((0.804296 + 3.332073 + 7.410494)/2) = 2.402797 pp.

Depth 3, Novel accuracy: mean = (71.000000 + 62.051282 + 47.487179)/3 = 60.179487; squared deviations = [117.083502, 3.503617, 161.094682]; s = sqrt((117.083502 + 3.503617 + 161.094682)/2) = 11.867641 pp.

Depth 3, Harmonic mean: mean = (78.189873 + 72.757513 + 60.511695)/3 = 70.486360; squared deviations = [59.344107, 5.158134, 99.493949]; s = sqrt((59.344107 + 5.158134 + 99.493949)/2) = 9.055280 pp.

Depth 3, Base macro-F1: mean = (86.700000 + 88.100000 + 83.000000)/3 = 85.933333; squared deviations = [0.587778, 4.694444, 8.604444]; s = sqrt((0.587778 + 4.694444 + 8.604444)/2) = 2.635021 pp.

Depth 3, Novel macro-F1: mean = (68.200000 + 57.400000 + 44.500000)/3 = 56.700000; squared deviations = [132.250000, 0.490000, 148.840000]; s = sqrt((132.250000 + 0.490000 + 148.840000)/2) = 11.865496 pp.

Depth 5, Base accuracy: mean = (77.595238 + 86.214286 + 90.666667)/3 = 84.825397; squared deviations = [52.275199, 1.929013, 34.120435]; s = sqrt((52.275199 + 1.929013 + 34.120435)/2) = 6.645474 pp.

Depth 5, Novel accuracy: mean = (48.102564 + 54.948718 + 59.538462)/3 = 54.196581; squared deviations = [37.137047, 0.565710, 28.535689]; s = sqrt((37.137047 + 0.565710 + 28.535689)/2) = 5.754930 pp.

Depth 5, Harmonic mean: mean = (59.388945 + 67.119066 + 71.877091)/3 = 66.128367; squared deviations = [45.419813, 0.981484, 33.047824]; s = sqrt((45.419813 + 0.981484 + 33.047824)/2) = 6.302742 pp.

Depth 5, Base macro-F1: mean = (75.700000 + 86.200000 + 90.700000)/3 = 84.200000; squared deviations = [72.250000, 4.000000, 42.250000]; s = sqrt((72.250000 + 4.000000 + 42.250000)/2) = 7.697402 pp.

Depth 5, Novel macro-F1: mean = (42.800000 + 51.100000 + 54.500000)/3 = 49.466667; squared deviations = [44.444444, 2.667778, 25.334444]; s = sqrt((44.444444 + 2.667778 + 25.334444)/2) = 6.018582 pp.

Depth 7, Base accuracy: mean = (92.190476 + 85.119048 + 93.928571)/3 = 90.412698; squared deviations = [3.160493, 28.022734, 12.361361]; s = sqrt((3.160493 + 28.022734 + 12.361361)/2) = 4.666079 pp.

Depth 7, Novel accuracy: mean = (71.897436 + 76.923077 + 73.538462)/3 = 74.119658; squared deviations = [4.938272, 7.859156, 0.337789]; s = sqrt((4.938272 + 7.859156 + 0.337789)/2) = 2.562735 pp.

Depth 7, Harmonic mean: mean = (80.789118 + 80.813789 + 82.492208)/3 = 81.365038; squared deviations = [0.331684, 0.303876, 1.270511]; s = sqrt((0.331684 + 0.303876 + 1.270511)/2) = 0.976236 pp.

Depth 7, Base macro-F1: mean = (92.200000 + 85.200000 + 94.000000)/3 = 90.466667; squared deviations = [3.004444, 27.737778, 12.484444]; s = sqrt((3.004444 + 27.737778 + 12.484444)/2) = 4.649014 pp.

Depth 7, Novel macro-F1: mean = (69.300000 + 76.200000 + 64.900000)/3 = 70.133333; squared deviations = [0.694444, 36.804444, 27.387778]; s = sqrt((0.694444 + 36.804444 + 27.387778)/2) = 5.695905 pp.

Depth 9, Base accuracy: mean = (93.880952 + 94.190476 + 92.666667)/3 = 93.579365; squared deviations = [0.090955, 0.373457, 0.833018]; s = sqrt((0.090955 + 0.373457 + 0.833018)/2) = 0.805428 pp.

Depth 9, Novel accuracy: mean = (68.717949 + 60.384615 + 63.435897)/3 = 64.179487; squared deviations = [20.597637, 14.401053, 0.552926]; s = sqrt((20.597637 + 14.401053 + 0.552926)/2) = 4.216137 pp.

Depth 9, Harmonic mean: mean = (79.352399 + 73.590843 + 75.314498)/3 = 76.085913; squared deviations = [10.669929, 6.225376, 0.595082]; s = sqrt((10.669929 + 6.225376 + 0.595082)/2) = 2.957227 pp.

Depth 9, Base macro-F1: mean = (93.700000 + 94.200000 + 92.700000)/3 = 93.533333; squared deviations = [0.027778, 0.444444, 0.694444]; s = sqrt((0.027778 + 0.444444 + 0.694444)/2) = 0.763763 pp.

Depth 9, Novel macro-F1: mean = (66.300000 + 54.700000 + 59.600000)/3 = 60.200000; squared deviations = [37.210000, 30.250000, 0.360000]; s = sqrt((37.210000 + 30.250000 + 0.360000)/2) = 5.823229 pp.

10. Parameter efficiency: derivation and measurements

Depth	Trainable	Total	Frozen	Trainable ratio (%)
1	395,008	124,718,849	124,323,841	0.3167
3	1,185,024	125,508,865	124,323,841	0.9442
5	1,975,040	126,298,881	124,323,841	1.5638
7	2,765,056	127,088,897	124,323,841	2.1757
9	3,555,072	127,878,913	124,323,841	2.7800

Post-freezing verification: P_train = sum(p.numel() for p in model.parameters() if p.requires_grad); P_total = sum(p.numel() for p in model.parameters()). Direct measurements match P_train(D)=395,008D and P_total(D)=124,323,841+395,008D. For depth 7: trainable ratio = 100×2,765,056/127,088,897 = 2.175687%. For depth 9: 100×3,555,072/127,878,913 = 2.780030%.

Depth 7 vs 9: trainable parameter reduction = (3,555,072−2,765,056)/3,555,072×100 = 22.222222%. Mean HM improvement = 81.365038−76.085913 = 5.279125 percentage points. Mean novel-accuracy improvement = 9.940171 pp. Mean base-accuracy difference = -3.166667 pp.

The descriptive metric HM per million trainable parameters equals HM/(P_train/1,000,000). This is not a universal efficiency score and tends to favor small parameter counts; it must be interpreted jointly with absolute accuracy and resource measures.

Depth	Seed	Base−novel gap (pp)	HM / million trainable

Depth	Seed	Base−novel gap (pp)	HM / million trainable
1	1	11.112	189.655
1	2	36.366	145.441
1	3	15.211	174.240
3	1	16.000	65.982
3	2	25.877	61.398
3	3	35.894	51.064
5	1	29.493	30.070
5	2	31.266	33.984
5	3	31.128	36.393
7	1	20.293	29.218
7	2	8.196	29.227
7	3	20.390	29.834
9	1	25.163	22.321
9	2	33.806	20.700
9	3	29.231	21.185

11. Figures and resource measurements

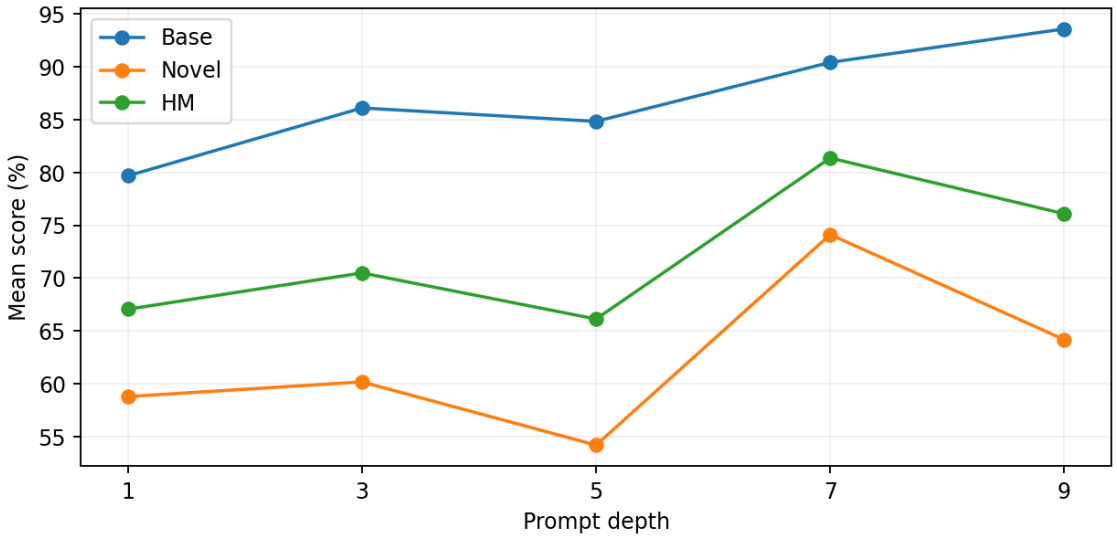


Figure 2. Historical depth-sweep plot: mean base accuracy, novel accuracy, and paired harmonic mean across three seeds, calculated from the supplied experimental logs.

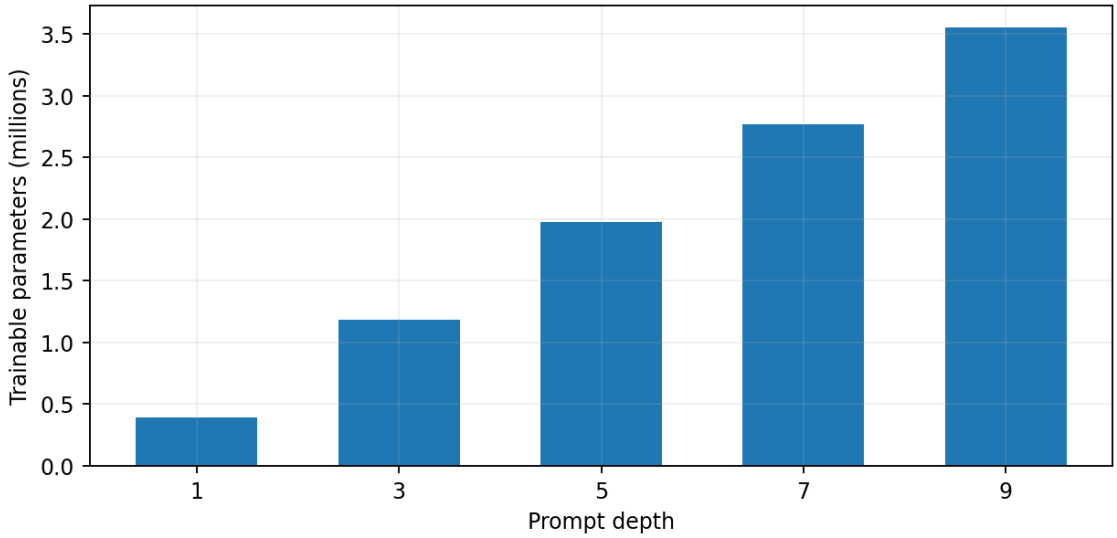


Figure 3. Historical parameter-efficiency plot: directly measured trainable-parameter count across depths (frozen CLIP encoders).

Depth	Recorded elapsed mean (s)	Sample SD (s)	Peak GPU memory
1	22.00	11.27	Not measured
3	14.67	0.58	Not measured
5	14.67	0.58	Not measured
7	14.33	0.58	Not measured
9	31.00	17.09	Not measured

Elapsed values were extracted from existing logs, but the logging scope and cold-start conditions are not sufficiently standardized to claim causal speed differences. Some depth-9 runs have notably longer elapsed times. Peak GPU memory was not present in the logs; it is neither zero nor estimated. A future controlled profiling protocol should synchronize CUDA, reset peak memory counters, and report consistent start/end boundaries.

12. Results, controls, uncertainty, and failure analysis

Observed results: depth 9 gives the highest base-class mean accuracy (93.579%) but depth 7 gives the highest novel-class mean accuracy (74.120%) and HM (81.365%). The depth-7 HM sample SD is 0.976 pp, compared with 2.957 pp at depth 9. This suggests a favorable accuracy/parameter tradeoff within the tested protocol, not evidence that depth 7 will always win on other datasets or under different training budgets. After implementing a separate deep-text-prompt penalty, the best observed K=9 regularized result is uniform gamma=0, lambda=0.1 (mean HM 79.828%). Applying that setting at K=7 yields 82.134% HM compared with the original K=7 baseline 81.365%; the improvement is small relative to seed variation.

Unfavorable examples must remain visible: depth 1 seed 2 achieves only 44.538% novel accuracy; depth 3 seed 3 achieves 47.487%; depth 5 seed 1 achieves 48.103%. These runs contribute to the

observed seed sensitivity. No individual seed was removed as an outlier. Three seeds are too few to establish broad statistical significance.

Controlled factors include backbone, class-split family, context-token count, shots, epochs, optimizer and the seed set. Prompt depth is the intervention. However, seed-specific few-shot training sets can vary, exact data hashes are missing, and a selected best depth evaluated on the same test set may benefit from model-selection bias. No paired statistical significance test is claimed. We also lack direct CoOp, CoCoOp and VPT training runs. The DAPR extension adds matched fixed-depth controls, gamma shape comparisons and lambda strength comparisons. Nonetheless lambda/depth preferences were influenced by test results, so the extension remains exploratory rather than a held-out confirmatory experiment.

Failure mechanism is interpretive, not directly diagnosed: increasing prompt capacity may improve adaptation to the five seen classes without improving unseen-class transfer. We did not inspect attention maps, representation geometry, or training/validation trajectories sufficiently to attribute the observed decline to overfitting conclusively.

The following figures expose seed-level variability, the accuracy–resource tradeoff, and the base-to-novel generalization gap. Error bars represent sample SD (n=3), not confidence intervals or statistical significance.

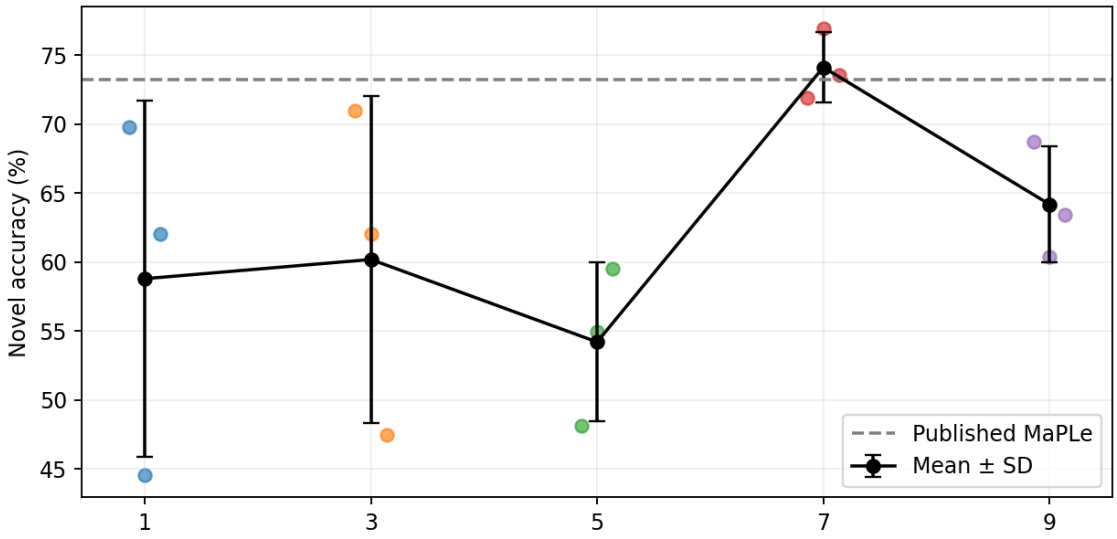


Figure 4. Individual seed novel-class accuracies and mean ± sample SD at each prompt depth. Horizontal dashed line is the published MaPLe EuroSAT novel score, provided only as context.

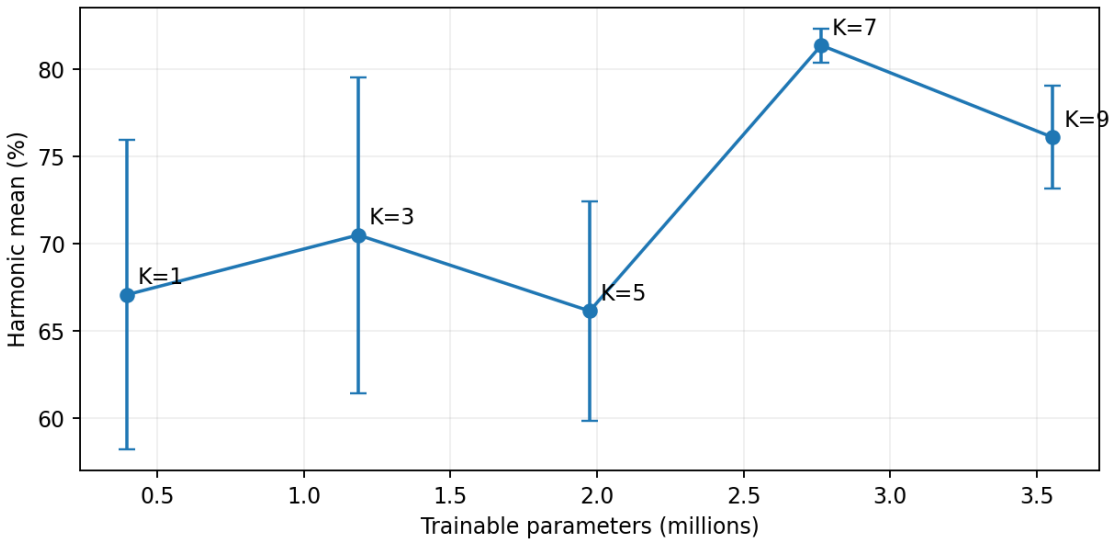


Figure 5. Parameter–performance tradeoff: mean seed-wise harmonic mean against trainable parameters, with sample-SD error bars. The depth-7 result is descriptive, not a validated optimum.

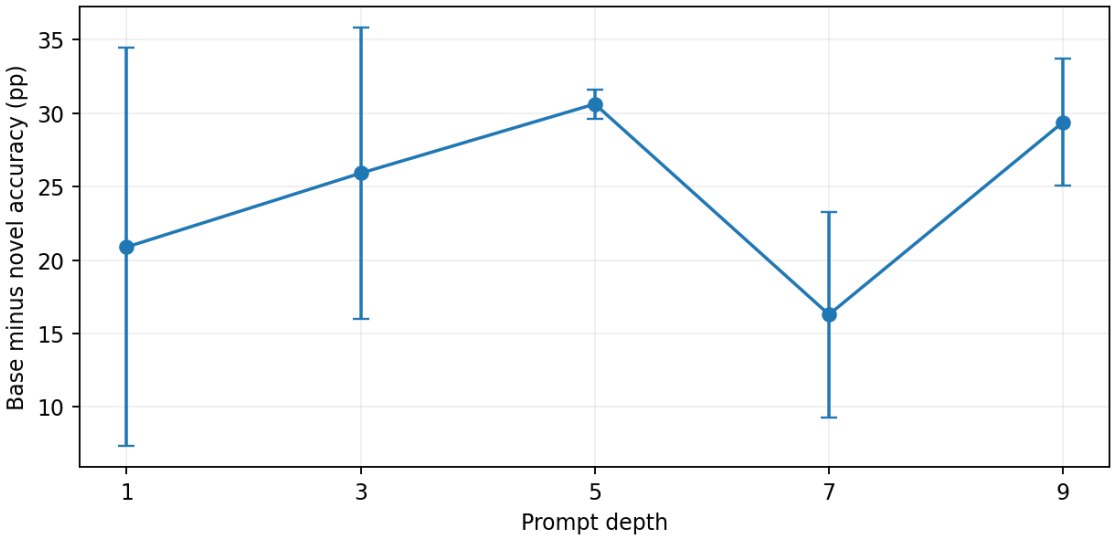


Figure 6. Mean base-minus-novel accuracy gap with sample-SD error bars. A smaller gap alone does not guarantee better task accuracy; interpret together with absolute base and novel scores.

Following the original prompt-depth sweep, we introduced DAPR: a targeted squared-L2 penalty on the deep text prompt tensors only. For D=K−1 compound text levels, set a_i=((i+1)/D)^γ and w_i=a_i / [(1/D)Σ_j a_j]. Then L_total = CE + λ[(1/D)Σ_i w_i mean(P_i²)]. Hence average(w_i)=1 for every gamma: γ=0 denotes uniform regularization, while γ=1 and γ=2 favor later levels. At λ=0 the extra term disappears. In the implementation, gradients still update prompt tensors and prompt-to-visual projections according to classification loss; the pretrained CLIP encoders remain frozen. No regularization is directly imposed on the visual projection matrices.

12.2 All DAPR and matched baseline aggregate results

Method	K	γ	λ	Mean Base %	Mean Novel %	Mean HM ± SD %
MaPLe baseline	9	—	0	93.579	64.180	76.086 ± 2.957
MaPLe baseline	7	—	0	—	74.120	81.365 ± 0.976
DAPR uniform	9	0	0.01	91.786	61.530	73.623 ± 1.451
DAPR uniform	9	0	0.1	94.071	69.752	79.828 ± 6.836
DAPR uniform	9	0	0.5	91.833	65.137	76.042 ± 3.402
DAPR linear	9	1	0.1	93.437	67.017	77.703 ± 7.113
DAPR quadratic	9	2	0.1	92.929	67.556	77.909 ± 6.253
DAPR uniform	7	0	0.1	92.778	73.744	82.134 ± 4.387

HM is calculated on each paired seed and then averaged. Standard deviations are sample SD (n=3). The original MaPLe K=7 baseline aggregate is preserved from the original 15-run report; the other prior K=9 aggregates correspond to previous experiment summaries. New Experiments 42–44 are independently recalculated from the supplied logs below. A difference between conditions is descriptive and is not automatically significant.

12.3 New log-backed experiments: raw counts and seed-wise audit

Exp. / setting	Seed	Base correct/total	Novel correct/total	Base %	Novel %	HM %
42: K9 λ0.01	1	3719/4200	2440/3900	88.548	62.564	73.322
42: K9 λ0.01	2	3957/4200	2290/3900	94.214	58.718	72.347
42: K9 λ0.01	3	3889/4200	2469/3900	92.595	63.308	75.201
43: K9 λ0.5	1	3753/4200	2809/3900	89.357	72.026	79.761
43: K9 λ0.5	2	3926/4200	2340/3900	93.476	60.000	73.087
43: K9	3	3892/4200	2472/3900	92.667	63.385	75.278

λ0.5						
44: K7 λ0.1	1	3971/4200	3104/3900	94.548	79.590	86.426
44: K7 λ0.1	2	3777/4200	2665/3900	89.929	68.333	77.658
44: K7 λ0.1	3	3942/4200	2859/3900	93.857	73.308	82.319

Example (Experiment 44, seed 1): Base = 3971/4200×100 = 94.547619%; Novel = 3104/3900×100 = 79.589744%; HM = 2×94.547619×79.589744/(94.547619+79.589744) = 86.426%. Other rows use the same direct-count arithmetic. The average of three seed-level HMs for Experiment 44 is 82.134% with sample SD 4.387 pp; Experiments 42 and 43 yield 73.623 ± 1.451 pp and 76.042 ± 3.402 pp respectively.

12.4 Gamma, lambda, and depth sensitivity

At K=9 with λ=0.1, uniform γ=0 achieves 79.828% HM, above γ=1 (77.703%) and γ=2 (77.909%). Thus the central hypothesis that a larger late-layer penalty necessarily improves novel-class transfer is not supported by these EuroSAT results. The small γ=2–γ=1 difference (~0.205 pp) is not evidence of an underlying systematic ordering. The λ sweep under γ=0 shows a non-monotonic trend: λ=0.01 → 73.623%; λ=0.1 → 79.828%; λ=0.5 → 76.042%. A weak or stronger penalty can be less effective than the evaluated middle setting; no continuous optimal λ is established.

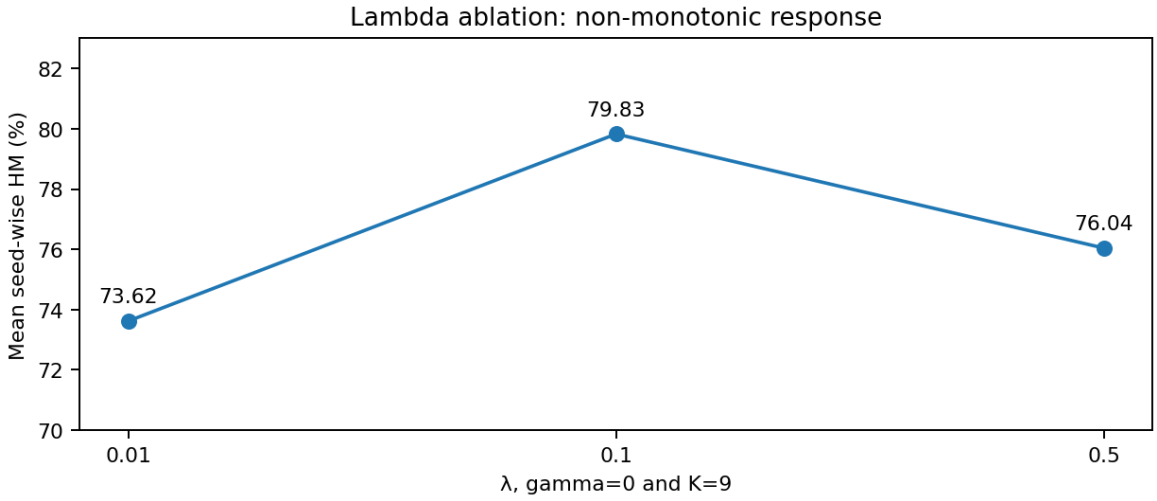


Figure 7. K=9 gamma=0 lambda ablation (mean of seed-wise HM). The plot connects three tested settings for visual legibility and does not imply measurements or continuity between them.

For a matched-depth comparison: MaPLe K=9 without DAPR has mean HM 76.086%; DAPR K=9 gamma=0 lambda=0.1 has 79.828% (+3.742 pp). At K=7, the baseline HM is 81.365% and regularized HM is 82.134% (+0.769 pp). DAPR K=7 exhibits seed-wise HMs 86.426%, 77.658%, and 82.319%; its HM sample SD is 4.387 pp. so the +0.769-pp mean difference should not be presented as a confirmed reliable effect.

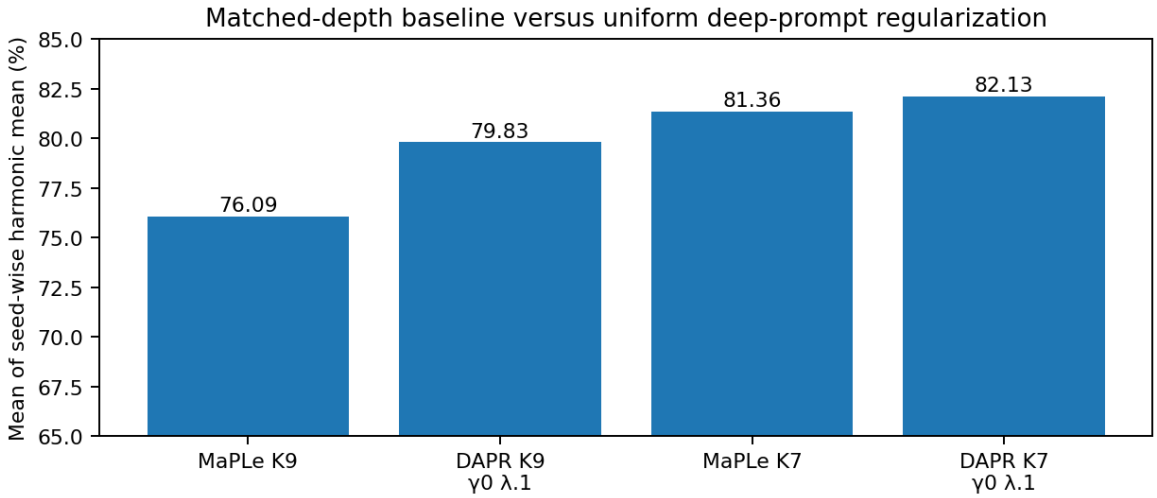


Figure 8. Fixed-depth comparisons at K=7 and K=9. Each value is an average of three seed-level harmonic means. Neither comparison corrects for test-guided selection.

12.5 Evidence, negative findings and limitations of the extension

The tested DAPR penalty is a genuine objective change relative to original MaPLe, but the experiments do not prove that depth-aware weights are better than uniform weights. In fact, $\gamma=0$ is the strongest of the three observed γ conditions at K=9. Regularization may influence optimization or reduce the magnitude of learned deep prompts, but neither mechanism is directly verified by parameter-norm trajectories, feature-space diagnostics, attention analysis, or paired predictions. We so report performance associations only, not a proven causal mechanism. Also, our selection of K=7 and $\lambda=0.1$ was informed by evaluation results on the same test partition; this invalidates confirmatory hyperparameter-selection claims. A separate validation regime is needed before a general improvement claim.

12.6 Reproducible DAPR commands and log manifest

Windows CMD (interactive; use %%s inside batch files): set "DAPR_LAMBDA=0.1" && set "DAPR_GAMMA=0" && (for %s in (1 2 3) do @(python train.py --root "E:\Codes\yzm5110\datasets" --seed %s --trainer MaPLe --dataset-config-file configs/datasets/eurosat.yaml --config-file configs/trainers/MaPLe/vit_b16_c2_ep5_batch4_2ctx.yaml --output-dir output/ablation/dapr_depth7_gamma0_lambda01/seed%s DATASET.NUM_SHOTS 16 DATASET.SUBSAMPLE_CLASSES base DATALOADER.NUM_WORKERS 0 TRAINER.MAPLE.PROMPT_DEPTH 7 && python train.py --root "E:\Codes\yzm5110\datasets" --seed %s --trainer MaPLe --dataset-config-file configs/datasets/eurosat.yaml --config-file configs/trainers/MaPLe/vit_b16_c2_ep5_batch4_2ctx.yaml --output-dir output/ablation/dapr_depth7_gamma0_lambda01/seed%s_novel_eval --model-dir output/ablation/dapr_depth7_gamma0_lambda01/seed%s --load-epoch 5 --eval-only DATASET.NUM_SHOTS 16 DATASET.SUBSAMPLE_CLASSES new DATALOADER.NUM_WORKERS 0 TRAINER.MAPLE.PROMPT_DEPTH 7)) > output_44_depth7_dapr_seed_123_base_novel_gamma_0_lambda_01.txt 2>&1

Uploaded log files for new experiments:
output_42_depth9_dapr_seed_123_base_novel_gamma_0_lambda_001.txt;

output_43_depth9_dapr_seed_123_base_novel_gamma_0_lambda_05.txt;
output_44_depth7_dapr_seed_123_base_novel_gamma_0_lambda_01.txt. Prior gamma=0/1/2 lambda=0.1 runs are represented by their experiment summaries and should be accompanied by full raw logs in the final repository. DAPR_LAMBDA and DAPR_GAMMA must be set explicitly for each run. The revised trainers/maple.py with normalized gamma weighting must also be supplied.

13. Limitations, conclusions, and unresolved questions

This study is limited to one satellite-image dataset (EuroSAT RGB), one pretrained backbone (CLIP ViT-B/16), 16-shot training, five epochs, three seeds, and a single prompt context length. DAPR adds a regularization term to deep text prompts, which is a targeted objective modification beyond the original depth hyperparameter sweep; it is not a new backbone architecture. Uniform gamma=0 outperforms the evaluated depth-dependent gamma=1 and gamma=2 settings at K=9, so the present evidence does not validate depth-aware weighting as superior. Hyperparameter/depth decisions were influenced by test results, creating selection bias. Missing elements include independent validation-based model selection, additional datasets, a public pinned repository, dataset/checkpoint checksums, standardized peak GPU memory, and paired per-image predictions. The reproduction gap to the published MaPLe EuroSAT novel result remains unresolved.

13.1 External benchmark discrepancy and reproducibility threats

The most important unresolved reproduction issue is a 9.051 pp local deficit in depth-9 novel-class accuracy relative to MaPLe's published EuroSAT result, despite a base-class difference of only 0.491 pp. A close base score does not establish full reproduction fidelity. The possible influence of stochastic few-shot samples, dataset/split artifacts, class-label text, preprocessing, pretrained checkpoints, software versions, and precision remains unresolved. These factors are competing hypotheses, not demonstrated causal explanations. Without source-commit, model-weight, and split-file hashes or an official-checkpoint replay, the discrepancy cannot be reliably assigned to the algorithm itself.

13.2 Uncertainty, selection bias, and resource limitations

Only three seeds were evaluated; novel-class scores change strongly among runs (depth-9 sample SD 4.216 pp, range 8.333 pp). The five depth conditions were judged on the reported test subsets. Since depth 7 was highlighted using these scores, the observed 5.279 pp mean-HM advantage over depth 9 is an exploratory within-protocol finding susceptible to test-based selection bias. No confidence interval, formal superiority test, or cross-dataset guarantee is claimed. The same local few-shot data and pipeline were intended across depth conditions, but exact artifact-level equality to the published benchmark is unverified.

Trainable parameter counts were read directly from PyTorch after CLIP freezing and are a valid parameter-efficiency measure for this implementation. They do not, alone, establish training latency, end-to-end energy efficiency, or GPU-memory savings. Recorded elapsed durations were not acquired under a standardized cold/warm-start profiling protocol, while peak CUDA memory was never recorded. as a result, parameter count is the principal defensible resource metric in the present report.

13.3 Interpretation and next tests

Our results show that deeper prompts do not always give better generalization. At depth 9, uniform regularization increased mean HM from 76.086% to 79.828%. At depth 7, it increased mean HM from

81.365% to 82.134%. However, depth-based weights did not perform better than uniform weights. These results describe our EuroSAT tests and should not be treated as universal findings.

The best observed setting in our EuroSAT experiments was depth 7 with uniform regularization (lambda = 0.1, gamma = 0). Its mean harmonic mean was 82.134%. The original MaPLe at depth 7 reached 81.365%. The improvement is small and is not enough to prove that the method is always better. Our results also do not show that depth-based weights are better than equal weights.

14. Reproducibility and runnable repository requirements

Local project root: E:\Codes\yzm5110\multimodal-prompt-learning. Dataset root: E:\Codes\yzm5110\datasets. Dataset config: configs/datasets/eurosat.yaml. Trainer config: configs/trainers/MaPLe/vit_b16_c2_ep5_batch4_2ctx.yaml. DAPR runs additionally require the edited trainers/maple.py with the normalized deep-text penalty; use environment variables DAPR_LAMBDA and DAPR_GAMMA. Experiment 44 checkpoints: output/ablation/dapr_depth7_gamma0_lambda01/seedN/MultiModalPromptLearner/model.pth.tar-5. Existing filenames/output paths retain historical naming conventions.

Example depth-7, seed-3, base-set checkpoint evaluation (Windows CMD):

python train.py --root "E:\Codes\yzm5110\datasets" --seed 3 --trainer MaPLe --dataset-config-file configs/datasets/eurosat.yaml --config-file configs/trainers/MaPLe/vit_b16_c2_ep5_batch4_2ctx.yaml --output-dir output/parameter_check/depth_7 --model-dir output/depth_study/depth_7/seed3 --load-epoch 5 --eval-only DATASET.NUM_SHOTS 16 DATASET.SUBSAMPLE_CLASSES base DATALOADER.NUM_WORKERS 0 TRAINER.MAPLE.PROMPT_DEPTH 7

For novel evaluation change DATASET.SUBSAMPLE_CLASSES to novel and point to the same depth/seed checkpoint. For depth 9, model checkpoints are under output/base2new/train_base/eurosat/shots_16/MaPLe/vit_b16_c2_ep5_batch4_2ctx/seedN; for other depths under output/depth_study/depth_D/seedN. These are evaluation commands, not fully verified training-reproduction scripts.

Project repository: https://github.com/siyahkut/yzm5110_research_1. The repository should include a README, dependencies, dataset instructions, configurations, training commands, result logs and scripts for the figures. Check that all files are available and record the final Git commit before submission.

15. Evidence provenance and calculation audit

Primary evidence supplied for this report: allzip.zip containing depth_study and base2new training/evaluation logs, local maple.py, and prompt-depth parameter-count logs. Derived data: maple_results_15_runs.csv and maple_summary_by_depth.csv. Correct counts and reported macro-F1 were extracted from evaluator logs; accuracy, harmonic mean, group statistics, ratios and gaps were calculated from the structured results. Original-paper published scores were not inferred from these local logs. DAPR evidence: user-supplied outputs for gamma=0 lambda=0.01 (Experiment 42), gamma=0 lambda=0.5 (Experiment 43), and gamma=0 lambda=0.1 K=7 (Experiment 44) are parsed from the explicit evaluator correct/total counts. Prior depth-9 gamma=0/1/2 lambda=0.1 summaries were verified in earlier log analysis and are reproduced as historical aggregates here. Exact seed-level

counts for those earlier conditions should be retained with the original logs in the eventual public repository.

A final evidence manifest should map each CSV row to its original log and checkpoint and record the dataset split file hash and Git commit. The supplied package does not establish all these provenance fields.

16. References

[1] Radford, A., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. ICML, PMLR 139, 8748–8763. https://proceedings.mlr.press/v139/radford21a.html

[2] Zhou, K., Yang, J., Loy, C. C., and Liu, Z. (2022). Learning to Prompt for Vision-Language Models. International Journal of Computer Vision, 130(9), 2337–2348. https://doi.org/10.1007/s11263-022-01653-1

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[4] Jia, M., et al. (2022). Visual Prompt Tuning. ECCV. https://www.ecva.net/papers/eccv_2022/papers_ECCV/html/4175_ECCV_2022_paper.php

[5] Khattak, M. U., Rasheed, H., Maaz, M., Khan, S., and Khan, F. S. (2023). MaPLe: Multi-Modal Prompt Learning. CVPR, 19113–19122. https://openaccess.thecvf.com/content/CVPR2023/html/Khattak_MaPLe_Multi-Modal_Prompt_Learning_CVPR_2023_paper.html

[6] Helber, P., Bischke, B., Dengel, A., and Borth, D. (2019). EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification. IEEE JSTARS, 12(7), 2217–2226.

[7] YZM5110 — Research Project assignment brief, pp. 1–3; 18 November 2026 deadline.

Primary source URLs are provided above; the MaPLe EuroSAT scores were verified against CVPR 2023 Table 3 (page 19119) and the average results against the same table. Any final paper-specific claims added by the student should be checked directly in the corresponding original works.

17. AI-use disclosure and integrity statement

ChatGPT (OpenAI) assisted with report structuring, English drafting/editing, literature comparison formatting, interpretation of DAPR design and limitations, exact numerical cross-checks from the supplied experiment logs, and charts. The student executed the local MaPLe/DAPR training and evaluations, edited/tested the local code, and supplied the outputs. The student remains responsible for verifying source papers, code, artifact provenance, computation, citations, and release compliance. This disclosure should be confirmed against the actual final workflow before submission.